IoT Value Chain Study Notes

100 % Based on the Professor's Slide

"The IoT Value Chain. Where would your business fit?"

⚡ **8 Components | Every platform name | Every category | All exam traps**

1. Slide Overview — The 8 Components

The slide shows the complete IoT value chain with 8 numbered components. Memorise their exact positions and what sits inside each one.

#	Component Name	Role in One Line
1	Sensors / Devices	Physical things that sense and send data
2	Wireless Sensor Networks (WSN)	Short-range wireless to move data to the gateway
3	Gateways + Network Servers	Bridge from short-range WSN to long-range carrier
4	Carrier Networks	Long-distance transport of data to the AEP
5	Application Enablement Platforms (AEP)	IoT middleware — where applications are built and run
6	Analytics Platforms	Over-the-top analytics on top of the AEP data
7	Device & Gateway Remote Management	Remote config and firmware updates (FOTA)
8	End-to-End System Integrators	Span the entire chain — deliver complete solutions

★ **Critical:** Value chain has 8 components. #5 = AEP, #6 = Analytics, #7 = Device Mgmt, #8 = SI.

⚠ **TRAP:** Standard web server is NOT one of the 8 components.

⚠ **TRAP:** Connectivity Management Platforms (CMPs) are INSIDE Component #4, not a separate top-level component.

2. Component #1 — Sensors / Devices

2.1 Device Makers (Hardware)

- Libelium
- Eurotech
- Telit
- Yanzi
- Witekio — embedded software ONLY (they do not make complete devices; they write firmware)

★ **Witekio:** Firmware development company only — they write embedded SW, they do NOT manufacture complete IoT devices.

2.2 Real-Time Operating Systems (RTOS) — from the slide

All of the following appear on the slide under Component #1:

RTOS	Type	Best For	MQTT?
Contiki-OS	Open Source	Resource-constrained IoT, IPv6, low power	Yes (MQTT-SN)
FreeRTOS	Open Source	Real-time embedded, microcontrollers	Yes (external lib)
Riot-OS	Open Source	Low-power IoT, modular, WSN	Yes (lightweight)
Zephyr	Open Source	Security-focused, multi-core, scalable	Yes (native)
TinyOS	Open Source	Wireless sensor networks (WSN) only	NO (no native MQTT)
OpenWrt	Open Source	GATEWAYS and routers — NOT small sensors	Yes (installable)

★ **All 6 RTOS:** ALL are open source. All support MQTT directly or via library EXCEPT TinyOS.
⚠ **TRAP:** OpenWrt is designed for GATEWAYS/routers, NOT for small IoT sensor devices.
⚠ **TRAP:** TinyOS has NO native MQTT support — only for wireless sensor networks.
⚠ **TRAP:** Not all IoT devices include RTOS — simple sensors may run bare-metal firmware.

2.3 Prototyping / Innovation Boards — from the slide

- Raspberry Pi
- Arduino
- Tilet
- Intel Edison
- MasterOfThings IoT Kit

★ **Purpose of prototyping boards:** Proof of concept, testing an idea, verifying/approving a project concept.

⚠ **TRAP:** Arduino and Raspberry Pi are NOT suitable for commercial IoT solutions by default — prototyping only!

⚠ **TRAP:** All IoT devices include a CPU/MCU chip — TRUE. All include RTOS — FALSE.

3. Component #2 — Wireless Sensor Networks

All technologies listed in this column are short-range or medium-range and mostly UNLICENSED (use ISM bands).

3.1 Every Technology on the Slide

Technology	Key Facts	Licensed?
RFID	Radio-Frequency ID. Passive tags get power from reader. Static data.	Unlicensed
NFC	Near Field Communication. ~1-10 cm. Passive, static data, powered by reader.	Unlicensed (13.56 MHz)
BlueTooth / BLE	BLE up to 300 m (BLE 5). Low power. 2.4 GHz ISM.	Unlicensed
BlueTooth Smart	Marketing name for BLE 4.0+.	Unlicensed
LoRa	Physical layer only. Up to 15 km. Cannot transfer video.	Unlicensed (868/915 MHz)
LoRaWAN	Network protocol over LoRa. NOT one global network. Roaming needed.	Unlicensed
MioTy	MIOTY — low power, high reliability. Used in Industry.	Unlicensed
MyThings (Behrtech)	Network platform for MIOTY, similar to LoRaWAN network server.	Unlicensed
Weightless (-P)	LPWAN standard. Unlicensed.	Unlicensed
Z-Wave	Smart home. Up to 100 m only.	Unlicensed
ZigBee	Mesh network. Up to 100 m. 2.4 GHz ISM.	Unlicensed
Xbee	ZigBee/802.15.4 module family.	Unlicensed
Ingenu (RPMA)	2.4 GHz ISM. Long range low power.	Unlicensed
PlatinusM2C	IoT connectivity solution.	Unlicensed
RIIM2	Radio in Infrastructure & Industry Mesh.	Unlicensed
MulteFire 1.1	LTE in unlicensed spectrum (Listen-Before-Talk).	Unlicensed
Wi-Sun	Smart utility metering mesh. IEEE 802.15.4g.	Unlicensed
Thread	IPv6 mesh for smart home. Based on 802.15.4.	Unlicensed
6LoWPAN	IPv6 over low-power wireless. Adapts IPv6 for 802.15.4.	Unlicensed
IQRF	Low power mesh, 868/916 MHz.	Unlicensed

★ **ISM bands:** UNLICENSED — free to use. 868 MHz = Europe/ME/Africa. 915 MHz = Americas. 2.4 GHz = Worldwide.

★ **WiFi uses:** 2.4 GHz ISM — same as BLE, ZigBee. But WiFi is NOT LPWAN and is NOT on this slide's WSN column.

⚠ **TRAP:** ZigBee, Ingenu, Z-Wave, Wi-Sun are ALL unlicensed — they all appear in slide column #2.

⚠ **TRAP:** LoRa and LoRaWAN CANNOT transfer video — far too low data rate.

⚠ **TRAP:** LoRaWAN is NOT one single global network — multiple independent networks, roaming IS needed.

4. Component #3 — Gateways & Network Servers

4.1 Gateway Hardware Makers — on the slide

- Intel (bold on slide — chipsets inside many gateways)
- Cisco
- Dell
- KerLink
- TEKTELIC
- Minew
- FaltCom (now Telia IoT Edge — industrial IoT gateway)
- Alleantia (now branded as Industrial Edge Platform)

★ **Gateway role:** Take data from short-range WSN and pass it to a long-range carrier network.

★ **Gateway CPU/RAM:** Must be sized relative to: number of devices + messages per device + offline disconnection time — ALL THREE.

⚠ **TRAP:** Gateways are the EDGE of the network — edge applications run ON the gateway.

4.2 LoRaWAN Network Servers — on the slide (listed inside Component #3)

- TTN — The Things Network (open-source, community-driven, global public network)
- Loriot (open-source, supports public/private/hybrid)
- ChirpStack (formerly LoRa Server, written in Go, open-source)
- LoRaServer (open-source, Go, supports LoRaWAN 1.0 and 1.1)
- MyThings (Behrtech — MIOTY central network platform)

★ **Network Server role:** Central authority for all gateways — decides which device can join which gateway. Gateways cannot decide alone.

⚠ **TRAP:** Cisco Jasper is NOT a LoRaWAN network server — it is a Connectivity Management Platform (CMP).

5. Component #4 — Carrier Networks

The slide breaks Component #4 into four sub-groups. Learn each name and which sub-group it belongs to.

5.1 LPWAN Operators — on the slide

- SigFox UnaBiz — one global network, acquired by UnaBiz

5.2 LEO Satellite Carriers — on the slide

- Lacuna — LoRaWAN satellite connectivity
- FossaSat 1u — pico-satellite, LEO
- Angels
- Hello
- Kineis 16U
- Swarm — LEO satellite (NOT a global SIM provider)

⚠ **TRAP:** ISS (International Space Station) is NOT listed and is NOT an IoT carrier — it is a research station.

⚠ **TRAP:** FossaSat is LEO, NOT GEO.

⚠ **TRAP:** Swarm is a satellite carrier — NOT a global SIM provider.

5.3 Global SIM / eSIM Providers — on the slide

- KORE
- OnoMondo
- Velos
- ARKESSA (Wireless Logic)
- Emnify
- THALES Adaptive Connect (formerly Gemalto)
- eseye

★ **Global SIM:** A SIM card with roaming pre-granted in most countries — solves the NB-IoT roaming problem for traveling devices.

⚠ **TRAP:** Emnify is a global SIM provider, NOT a LEO satellite carrier.

⚠ **TRAP:** Lacuna is a satellite LoRaWAN provider, NOT a global SIM provider.

⚠ **TRAP:** Both Arkessa AND Swarm are NOT both global SIM providers — Arkessa yes, Swarm is satellite.

5.4 Connectivity Management Platforms (CMPs) — on the slide

CMP Platform	Parent Company / Notes
Jasper CC (Cisco Control Center)	Originally Jasper Wireless — first CMP ever built, acquired by Cisco
Ericsson DCP / Accelerator (Aries)	Built by Ericsson with billing capabilities, later sold to Aeris

Vodafone GDSP (→ Vodafone IoT)	2024: Vodafone moved it to a joint venture with Microsoft
Verizon ThingSpace	Verizon's own CMP, US-based
Nokia WING	Nokia Worldwide IoT Network Grid, started 2017
IoTM (Israel)	Over-the-top (OTT) CMP solution

★ **CMP purpose:** Enable OR disable each device's connectivity based on defined conditions. Handles billing and SIM management.
★ **First CMP ever:** Jasper Wireless — now Cisco Control Center.
⚠ **TRAP:** CMP does NOT monitor the content of the data — only the connectivity itself.

5.5 Cellular Network Technologies — on the slide

- GPRS (2G)
- 3G
- 4G LTE variants: LTE-M, NB-IoT (NB LTE), LTE Lite, Cellular IoT, MulteFire LTE
- 5G Massive IoT (mMTC — massive Machine Type Communications)

★ **NB-IoT:** Bidirectional. Supports TCP/IP AND UDP. MQTT IS supported. Main challenge = roaming.
★ **3G / 4G:** Drain IoT device battery — too power-hungry for most LPWAN IoT use cases.
⚠ **TRAP:** NB-IoT is bidirectional — FALSE is the wrong answer here. It IS bidirectional.

6. Component #5 — Application Enablement Platforms (AEP)

This is the most tested component. The slide lists every AEP platform — learn ALL of them and classify them correctly.

6.1 Every AEP Platform on the Slide

Platform	Owner / Notes	Sub-category
MasterOfThings	SpimeSenseLabs — professor's own platform, featured prominently	Licensed productivity AEP
VoT / B-Scada	Listed under AEP on the slide (but exam may classify as Analytics — see trap)	AEP / SCADA
ThingWorx IIoT	PTC — industrial IoT AEP	Licensed IIoT AEP
Xively	Google GCP — was one of the first IoT AEPs	Licensed AEP
Cisco Kinetc	Cisco's IoT AEP offering	Licensed AEP
ThingsBoard	Open-source, Ukrainian company	Open Source AEP
FiWare	Open-source — EU initiative	Open Source AEP
CityOS	Smart city focused AEP	Open Source AEP
Gimalto / Gemalto	Also appears as connectivity — context dependent	AEP / Connectivity
Siemens Mindsphere	Siemens industrial IoT AEP	IIoT AEP
Schneider WonderWare	Schneider Electric industrial SCADA/AEP	IIoT AEP
Cumulocity IIoT	Software AG — horizontal IoT AEP	Licensed AEP
GE Predix IIoT	General Electric industrial IoT AEP	IIoT AEP
PlatOne (SAP Leonardo)	SAP's IoT platform	Licensed IIoT AEP
Carriots (Altair)	IoT AEP, acquired by Altair	Licensed AEP

⚠ **TRAP:** B-Scada: The slide places it under AEP — but the professor's own exam (Q42) classifies it as Analytics. Answer: ANALYTICS in the exam.

⚠ **TRAP:** FiWare is an open-source AEP — NOT a device management platform.

⚠ **TRAP:** Gemalto/Gimalto can appear as AEP or SIM provider — context matters.

6.2 The 4 Mandatory AEP Modules (Core Knowledge)

Mod	Name	Key Functions	Common Trap

ul e			
A	Sensor / Device Profile Definition	Define sensors, drivers, data types, protocols, import/export devices	Does NOT collect data
B	Monitoring, Enrichment & Orchestration	Real-time monitoring, rule-based engine, missing data alarms, data correlation, trigger actions	Includes rule engine + alarms
C	Users Access Rights Management	Manage developers AND end-users access to BOTH apps AND devices	Manages device access too, not just apps
D	Visual IDE (Development Environment)	Drag-and-drop, code window, frontend+backend, real-time preview	MOST IMPORTANT module for developers

★ **AEP module count:** Exactly 4 mandatory modules — no more, no less.
★ **80% rule:** ~80% of every IoT solution is pre-built common functionality provided by the AEP.
⚠ **TRAP:** Device Profile Definition does NOT collect data — that is Module B.
⚠ **TRAP:** AEP is NOT a framework, NOT a software library, NOT a development stack.

6.3 Optional App Modules Built on top of AEP

- Alarm Management — map-based alarm tracking for sensors/sites
- Task Management — assign tasks to field workforce
- Asset Management — manage physical assets on a map
- Global Calendar — manage teams, events, vacations

⚠ **TRAP:** These are OPTIONAL APPLICATION MODULES, not mandatory AEP core modules.

7. Component #6 — Analytics Platforms (Over-OTT)

The slide shows "OverOTT" as a category label for this column — all analytics platforms sit on top of the AEP as Over-The-Top solutions.

Every Analytics Platform on the Slide

- Splunk
- Grok
- Hadoop
- Watson IoT (IBM)
- OneMind

★ **Hadoop:** An analytics/big data platform — NOT an IoT platform. It is listed under Analytics (Component #6) on the slide.

⚠ **TRAP:** Hadoop is an Analytics platform, not an IoT AEP.

⚠ **TRAP:** IBM Watson IoT is an Analytics platform in this slide's classification.

8. Component #7 — Device & Gateway Remote Management

Component #7 is a horizontal layer spanning from Component #1 to Component #4 on the slide.

Every Platform on the Slide

- Balena
- Kaa
- Bosch IoT Suite (highlighted in red on the slide)
- Nokia IMPACT (highlighted in red on the slide)

★ **Device Mgmt purpose:** Remote device CONFIGURATION + Remote FIRMWARE UPDATES (FOTA — Firmware Over The Air). That is ALL it does.

⚠ **TRAP:** Device Management does NOT collect sensor data — that is the AEP's role.

⚠ **TRAP:** Bosch IoT Suite = Device Management (#7), NOT an AEP (#5).

⚠ **TRAP:** Nokia IMPACT = Device Management (#7), NOT a CMP and NOT an AEP.

9. Component #8 — End-to-End System Integrators

The slide shows Component #8 as a long arrow spanning across all 6 main components (1–6). System Integrators design, build, and deliver complete IoT solutions end-to-end.

★ **Role:** End-to-End System Integrators span the entire value chain and are responsible for delivering complete solutions.

★ **Job title:** The IoT Solution Architect is the person who designs end-to-end IoT solutions. Embedded System Engineers only work on Component #1.

⚠ **TRAP:** Embedded System Engineer = devices only. IoT Integration Engineer = integrates systems, does NOT develop devices.

10. Master Platform Classification Table — Memorise Every Row

This table covers every platform name visible on the slide. Be able to classify each one instantly.

Platform Name	Component #	Category
Libelium / Eurotech / Telit / Yanzi	#1	Sensor/Device Makers
Witekio	#1	Embedded SW (firmware only, NOT device maker)
Contiki-OS / FreeRTOS / Riot-OS / Zephyr / TinyOS / OpenWrt	#1	RTOS for IoT devices
Raspberry Pi / Arduino / Intel Edison / MoT Kit / Tilet	#1	Prototyping/Innovation Boards
RFID / NFC / BLE / LoRa / LoRaWAN / ZigBee / Z-Wave / Ingenu / Wi-Sun / Thread / 6LoWPAN / MioTy / Weightless / MulteFire / IQRF	#2	Wireless Sensor Network Technologies
Intel / Cisco / Dell / KerLink / TEKTELIC / FaltCom / Alleantia / Minew	#3	Gateway Hardware Makers
TTN / Loriot / ChirpStack / LoRaServer / MyThings	#3	LoRaWAN Network Servers
SigFox UnaBiz	#4	Carrier Network (LPWAN Operator)
Lacuna / FossaSat / Angels / Hello / Kineis / Swarm	#4	Carrier Network (LEO Satellite)
KORE / OnoMondo / Velos / Arkessa / Emnify / THALES / eseye	#4	Carrier Network (Global SIM Provider)
Jasper CC (Cisco) / Ericsson DCP / Vodafone GDSP / Verizon ThingSpace / Nokia WING / IoTM	#4	Carrier Network (CMP — Connectivity Mgmt Platform)
GPRS / 3G / 4G LTE / NB-IoT / LTE-M / 5G mMTC	#4	Carrier Network (Cellular Technology)
MasterOfThings	#5	Application Enablement Platform (AEP)
VoT / B-Scada	#5 (slide) / #6 (exam)	AEP on slide BUT answer Analytics in exam
ThingWorx IIoT (PTC)	#5	Application Enablement Platform (AEP)
Xively (Google GCP)	#5	Application Enablement Platform (AEP)
Cisco Kinetc	#5	Application Enablement Platform (AEP)
ThingsBoard	#5	Application Enablement Platform (AEP — Open Source)

FiWare	#5	Application Enablement Platform (AEP — Open Source, NOT device mgmt)
CityOS	#5	Application Enablement Platform (AEP)
Gimalto / Gemalto	#5 / #4	AEP or Global SIM — context dependent
Siemens Mindsphere	#5	Application Enablement Platform (IIoT AEP)
Schneider WonderWare	#5	Application Enablement Platform (IIoT AEP)
Cumulocity IIoT	#5	Application Enablement Platform (AEP)
GE Predix IIoT	#5	Application Enablement Platform (IIoT AEP)
PlatOne (SAP Leonardo)	#5	Application Enablement Platform (AEP)
Carriots (Altair)	#5	Application Enablement Platform (AEP)
Splunk / Grok / Hadoop / Watson IoT / OneMind	#6	Analytics Platform (Over-OTT)
Balena / Kaa / Bosch IoT Suite / Nokia IMPACT	#7	Device & Gateway Remote Management
End-to-End System Integrators	#8	Cross-chain component spanning #1–#6

11. SigFox — The Most Tested Topic

SigFox appears in Component #4 on the slide as 'SigFox UnaBiz'. More exam questions come from SigFox than any other single technology.

SigFox Fact	Value / Answer
Max bytes per uplink message	12 bytes
Max bytes per downlink message	8 bytes
Max uplink messages per day	140 messages
Max bytes per DAY (uplink)	140 × 12 = 1,680 bytes
Max bytes per DAY (downlink)	4 × 8 = 32 bytes
Is it one global network?	YES — no roaming subscription needed
Protocol stack	NOT TCP/IP based
MQTT support?	NO — MQTT needs TCP/IP, SigFox does not use it
Can it run as a private network?	NO — licensed carrier infrastructure only
Can an individual buy a SigFox BTS?	NO — carrier infrastructure, not sold to individuals
Can it transfer video?	NO — far too low data rate
Frequency band	Unlicensed ISM — 868 MHz (Europe/ME/Africa) or 915 MHz (Americas)
Who acquired SigFox?	UnaBiz — now called SigFox UnaBiz
What did SigFox competition force?	Pushed 3GPP to develop NB-IoT as a low-power alternative

12. IoT Job Roles — From the Slide's Architecture

Job Role	Works On	Common Exam Trap
IoT Solution Architect	Designs the ENTIRE end-to-end IoT solution (all 8 components)	NOT responsible for developing devices — that is the embedded engineer
Embedded System Engineer	Component #1 ONLY — device HW design and firmware development	Does NOT design end-to-end solutions
IoT Integration Engineer / SE	Integrates platforms across the chain	Does NOT develop IoT devices
IoT Platform Developer	Works on Component #5 (AEP)	Develops the platform itself
IoT Application Developer	Builds apps running on top of Component #5	Uses the AEP's Visual IDE

13. Last-Minute Exam Traps — All From This One Slide

13.1 Platform Classification Traps

⚠ **TRAP:** FiWare = AEP (#5) — NOT device management (#7).
⚠ **TRAP:** Hadoop = Analytics (#6) — NOT an IoT platform.
⚠ **TRAP:** Balena / Kaa = Device Mgmt (#7) — NOT AEP (#5).
⚠ **TRAP:** Bosch IoT Suite = Device Mgmt (#7) — NOT AEP (#5).
⚠ **TRAP:** Nokia IMPACT = Device Mgmt (#7) — NOT a CMP.
⚠ **TRAP:** B-Scada: slide says AEP, but exam answer = Analytics.
⚠ **TRAP:** Cisco Kinetc = AEP (#5). Cisco Jasper = CMP (#4). Know the difference!
⚠ **TRAP:** Nokia WING = CMP (#4). Nokia IMPACT = Device Mgmt (#7). Different products!
⚠ **TRAP:** THALES (Adaptive Connect) = Global SIM provider (#4). NOT an AEP.
⚠ **TRAP:** Swarm = LEO satellite carrier (#4). NOT a global SIM provider.
⚠ **TRAP:** Emnify = Global SIM provider (#4). NOT a satellite carrier.
⚠ **TRAP:** Lacuna = satellite LoRaWAN (#4). NOT a global SIM provider.
⚠ **TRAP:** ISS = NOT an IoT satellite. Not on the slide.
⚠ **TRAP:** FossaSat = LEO (not GEO).

13.2 Wireless Technology Traps

⚠ **TRAP:** SigFox = 12 bytes/msg max. NOT 24 bytes.
⚠ **TRAP:** SigFox = 1680 bytes/day max (uplink). NOT 'unlimited'.
⚠ **TRAP:** SigFox = NOT TCP/IP → NO MQTT.
⚠ **TRAP:** SigFox = ONE global network → NO roaming needed.
⚠ **TRAP:** SigFox CANNOT operate as a private network.
⚠ **TRAP:** LoRaWAN ≠ one global network → roaming IS needed.
⚠ **TRAP:** LoRa and LoRaWAN CANNOT transfer video.
⚠ **TRAP:** NB-IoT IS bidirectional (both upload and download).
⚠ **TRAP:** NB-IoT DOES support MQTT (TCP/IP + UDP).
⚠ **TRAP:** NB-IoT main challenge = ROAMING.
⚠ **TRAP:** 3G drains IoT device battery — use NB-IoT/LoRa instead.
⚠ **TRAP:** ISM band = UNLICENSED. Not exclusive to IoT.
⚠ **TRAP:** 868 MHz = Europe, ME, Africa. 915 MHz = Americas.
⚠ **TRAP:** WiFi and BLE both use 2.4 GHz ISM band.
⚠ **TRAP:** WiFi is NOT an LPWAN technology.
⚠ **TRAP:** ZigBee, Ingenu, Z-Wave, Wi-Sun = ALL unlicensed (ISM).
⚠ **TRAP:** OpenWrt = for GATEWAYS, NOT for small IoT sensors.
⚠ **TRAP:** TinyOS = NO native MQTT support.

13.3 Architecture & Role Traps

⚠ **TRAP:** Value chain = 8 components (not 6).
⚠ **TRAP:** AEP = exactly 4 mandatory modules.
⚠ **TRAP:** Applications CANNOT talk directly to devices — AEP is the mandatory middle layer.
⚠ **TRAP:** Carrier networks carry data from GATEWAY (or device) to AEP — NOT from device to gateway.

⚠ **TRAP:** Device-to-Gateway = WSN (Component #2). Gateway-to-AEP = Carrier (Component #4).
⚠ **TRAP:** Arduino/Raspberry Pi = prototyping ONLY, not commercial.
⚠ **TRAP:** Video cameras = NOT standard IoT devices (too much data for AEP).
⚠ **TRAP:** AEP cannot process video feeds.
⚠ **TRAP:** Solution Architect = end-to-end design. Embedded Engineer = devices only.
⚠ **TRAP:** IoT Integration Engineer does NOT develop IoT devices.
⚠ **TRAP:** Gateway CPU/RAM depends on ALL: device count + msg rate + disconnection time.
⚠ **TRAP:** MQTT = ISO/IEC 20922 standard. TCP/IP based. Lightweight + bidirectional + for constrained devices.
⚠ **TRAP:** CoAP = legacy protocol. No longer widely used.
⚠ **TRAP:** HTTP = NOT a good IoT protocol (too heavy).

YOU GOT THIS. GO SMASH THAT EXAM!